

Hallucinogenic Drugs Attenuate the Subjective Response to Alcohol in Humans

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This study investigated possible interactions between alcohol and hallucinogens in 22 lysergic acid diethylamide (LSD) and/or psilocybin users through retrospective structured interviews. Of those who had used LSD with alcohol, 86.7 per cent reported a complete blockade of subjective alcohol effects, while the remaining cases reported a diminished response. In addition, 60 per cent of respondents who had used alcohol and psilocybin together reported a partial antagonism of subjective alcohol effects. *T*-test analyses revealed that LSD's antagonism of alcohol effects were significantly greater than those associated with psilocybin. It is proposed that LSD's effect on alcohol intoxication may involve interactions with various serotonergic and/or dopaminergic receptor systems. Copyright © 2000 John Wiley & Sons, Ltd.

KEY WORDS — lysergic acid diethylamide; alcohol; psilocybin; hallucinogens; serotonin; dopamine

INTRODUCTION

The therapeutic use of lysergic acid diethylamide (LSD) has been studied extensively as a possible adjunct to psychotherapy for the treatment of alcoholism (e.g. Abraham *et al.*, 1996; Mangini, 1998). Such studies have produced inconsistent results and this area of research was abandoned in the early 1970s due to concerns over LSD's safety (Mangini, 1998). Despite the fact that several clinical trials were conducted over a 25-year period, the effects of the combined use of alcohol and LSD were never elucidated. Nevertheless, the pharmacological profiles of these two substances suggest the possibility of a clinically significant interaction. For example, LSD binds with high affinity at both 5-HT₁ and 5-HT₂ receptors (e.g. Peroutka, 1994), and drugs that act on either of these receptor subtypes have been shown to alter both ethanol intake and discriminative stimulus effects in rats (Szeliga and Grant, 1998; Kostowski and Bienkowski, 1999). Furthermore, LSD is thought to act as a

partial agonist at dopamine (DA) receptors (Watts *et al.*, 1995; Giacomelli *et al.*, 1998), and mid-brain DA transmission has been implicated in several of alcohol's reinforcing properties (e.g. Koob *et al.*, 1998). Given the fact that LSD is known to affect neural systems implicated in ethanol intake, reinforcement, and discriminative stimulus, one might expect the subjective human response to ethanol to be altered by LSD administration. Interestingly, a recent study of the drug taking patterns of adolescent drinkers suggests that the simultaneous use of alcohol and hallucinogens is quite common, with 16 per cent of the sample reporting combining alcohol with hallucinogens on at least one occasion (Martin *et al.*, 1996). The current investigation presents systematically collected information from individuals who have reported using ethanol with LSD and/or psychotic mushrooms, a related hallucinogenic drug.

METHODS

Participants for this retrospective self-report study were recruited through the 'snowball' method of sampling. This sampling method has been widely used to study hidden or clandestine populations

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(Jackson, 1997), and requires the researcher to make an initial set of contacts and to ask these contacts to introduce him/her to potential participants. In the present study four contacts whose past hallucinogenic use was known *a priori* were used to recruit subjects. Potential participants were contacted either in person or by telephone and were asked if they would answer some questions about their alcohol and drug use for a scientific study. Of the 25 individuals contacted, three declined to participate, leaving a final sample of 22 subjects.

Structured interviews were conducted orally and recorded via audio-tape. Participants were assured of their anonymity and informed that their audio-taped responses would remain strictly confidential. The interview consisted of a series of standardized open-ended questions designed to elicit information about the participant's previous experiences with alcohol, LSD, and psilocybic mushrooms when used alone or in combination. In order to avoid any possible investigator influence, all questions about possible alcohol/hallucinogenic interactions were always phrased as follows: have you ever used alcohol while under the influence of LSD (magic mushrooms) or LSD (magic mushrooms) while under the influence of alcohol? Did you notice any difference in the effect of either the alcohol or the LSD (magic mushrooms) when you used the two together? What was the nature of the difference? Can you give me a specific example about a time when this happened? In order for a report to be considered usable, the subject had to have used at least the minimum dose required to detect the subjective effects of each drug when used alone.

Reported drug interaction effects were rated by the principal investigator and by an independent rater who was blind to the study's hypotheses. Discrepancies in the coding were resolved by having a second independent rater score any interviews in question. Using a five-point Likert scale (1 = completely blocked, 2 = somewhat diminished, 3 = neutral, 4 = somewhat enhanced, 5 = strongly enhanced), the raters first scored the reported effect of alcohol when it was taken with LSD. Using the same Likert scale, the raters then coded the reported effect of LSD when it was taken in conjunction with alcohol. Ratings of reported effects of alcohol and psilocybin when the two drugs were used in combination were assessed in the same manner. Raters were instructed to eliminate cases in which interviewer bias was present. A complete transcript of all interviews is available from the authors upon request.

RESULTS

Nineteen males and three females were interviewed for the present study. The participants ranged in age from 18 to 28 years ($M = 23.73$, $SD = 2.86$). Most participants reported extensive past use of hallucinogens, with 73 per cent of subjects reporting 20 or more lifetime uses of LSD and 73 per cent reporting 20 or more lifetime uses of psilocybin. Drug interaction effects were not analysed for four subjects who reported that they had never used intoxicating levels of alcohol in combination with either LSD or psilocybin and for one participant who provided inconsistent, and thus uncodable, responses to interview questions. Only partial effects were analysed for two subjects who reported never having used intoxicating levels of alcohol with LSD, and for two subjects who reported never having used intoxicating levels of alcohol with psilocybin.

Pearson correlation coefficients were calculated in order to determine the inter-rater reliabilities for the coding of the interviews. These analyses revealed a very high concordance between raters for both LSD-alcohol interactions ($r = 0.985$, $p < 0.001$) and psilocybin-alcohol interactions ($r = 0.871$, $p < 0.001$). An analysis of the interview ratings revealed a striking antagonism of subjective alcohol effects when it was used in combination with LSD (see Table 1). Specifically, 86.7 per cent of the sample reported experiencing a complete blockade of subjective alcohol effects, while the remainder reported a diminished effect. No differences in subjective LSD effects were associated with alcohol use in any of the subjects. There was also a weaker tendency for subjects to report a diminished effect of alcohol when used in conjunction with psilocybic mushrooms, with 60 per cent of the sample noting a diminished effect, 34 per cent reporting no change and 6.7 per cent reporting an enhanced effect. The majority of the sample (80 per cent) also reported no change in psilocybin's effects when it was used with alcohol. However, two subjects reported a diminished effect, and one subject reported an enhanced effect of psilocybin when used in combination with alcohol. In addition, one participant voluntarily reported experiencing a pleasant synergistic effect when taking psilocybin with alcohol, while another participant reported having a very unpleasant synergistic effect.

In order to compare the effects of LSD and psilocybin on the subjective response to alcohol, a paired samples *t*-test was performed. This analysis revealed that LSD's antagonism of the alcohol

Table 1. Subjective effects associated with concurrent use of alcohol and hallucinogens

Id	Age	Sex	Estimated lifetime use of LSD	Estimated lifetime use of psilocybin	Concurrent use of alcohol and LSD		Concurrent use of alcohol and psilocybin	
					Subjective alcohol effects	Subjective LSD effects	Subjective alcohol effects	Subjective psilocybin effects
A	21	M	100–150	50–100	blocked	no change	no change	no change
B	26	M	20	20	blocked	no change	no change	no change
C	21	M	50	40–50	–	–	diminished	diminished
G	25	M	60–90	60–90	blocked	no change	diminished	no change
H	27	F	60–75	30–35	blocked	no change	–	–
I	28	M	30–50	20–30	blocked	no change	diminished	no change
J	24	M	25	50	blocked	no change	no change	no change
K	28	M	10–25	10–25	blocked	no change	diminished	no change
L	25	M	35	50	diminished	no change	diminished	no change
M	24	M	40	50	blocked	no change	no change	no change
P	24	M	> 50	30–40	–	–	diminished	no change
R	24	F	30	2	blocked	no change	–	–
S	23	F	3	30	blocked	no change	no change	diminished
T	24	M	25	30	blocked	no change	enhanced	enhanced
U	26	M	200–300	50	blocked	no change	diminished	no change
V	28	M	25	50	blocked	no change	diminished	no change
W	22	M	15	30–35	diminished	no change	diminished	no change

effect is significantly greater than that associated with psilocybin $t(12) = 5.741, p < 0.001$.

REPRESENTATIVE SUBJECTIVE REPORTS

LSD–alcohol interactions

G, a 25-year-old male, reported extensive experience with LSD, using it on 60–90 separate occasions. Typically, G can begin to discern the effects of alcohol after four drinks and it takes approximately seven or eight drinks for G to become significantly intoxicated. According to G, 'When I was on LSD I found that no matter what quantity of alcohol I consumed on the high, I did not notice any of the effects of the alcohol'. G recalled one specific occasion when he drank approximately 12 beers over the course of an evening after ingesting LSD: '... at that point, I didn't notice the effects of the alcohol at all. I was not drunk, nothing. I just felt a little bloated.'

H, a 27-year-old female, reported extensive experience with LSD, using it on between 60 and 75 separate occasions. She reported using an intoxicating dose of alcohol (more than seven drinks) in combination with LSD 'several' times. According

to H, LSD–alcohol interactions work two ways. 'The first example would be if I were to use LSD and then drink, I wouldn't feel any of the effects of alcohol until after the effects of the LSD had worn off ... and if I were to take LSD once I was already subjectively feeling drunk, the alcohol effects would go away after taking the LSD.'

T, a 24-year-old male, reported using LSD on 25 occasions. T reported that there were several occasions where he consumed an intoxicating dose of alcohol (more than six drinks) while under the influence of LSD. According to T, 'You can't get drunk when you are on LSD ... One time I drank about 18 beers, a half bottle of Bailey's and a Mickey of vodka on two hits of acid and I didn't feel anything from the booze ... it was just a normal LSD high.'

Psilocybin–alcohol interactions

B, a 26-year-old male, has used psilocybic mushrooms on 20 separate occasions. When using alcohol concurrently with psilocybic mushrooms, B claims that 'I could feel a combination of both. It is not like LSD where the alcohol effects are blocked. Sometimes one would be stronger than the other,

but that would just depend on how much I took of each'.

L, a 25-year-old male, has used psilocybic mushrooms on 50 separate occasions. According to L, when using psilocybin with a normally intoxicating dose of alcohol (more than four drinks), '... it produces a more mellow buzz. The alcohol effect was deadened'. When asked to compare this experience with that of LSD, 'I could feel the alcohol more for sure on mushrooms'.

V, a 28-year-old male, reported 50 previous uses of psilocybic mushrooms, including several occasions where he consumed a normally intoxicating dose of alcohol (more than six drinks). According to V, when using alcohol and psilocybic mushrooms together: 'You can tell that you've drunk alcohol when on mushrooms, but there would be less of an effect ... you could feel it more than when on LSD'.

DISCUSSION

The present data indicate that the subjective response to ethanol is strongly attenuated by the administration of LSD. Of the 15 LSD users who reported concurrent use of alcohol, 13 reported a complete blockade of alcohol's subjective effects, while the remaining two cases reported experiencing a virtual elimination of subjective alcohol effects when the drug was combined with LSD. The finding that LSD's antagonism of alcohol's effects reportedly occurred irrespective of whether the subject had consumed LSD prior to drinking or the LSD was ingested when already intoxicated further supports the conjecture that LSD's effect on the subjective response to alcohol may be due to a direct pharmacological interaction. In contrast to LSD, the use of psilocybic mushrooms failed to fully antagonize the subjective effects of alcohol. Although nine of the 15 psilocybin users reported a diminished response to alcohol, no participants reported that the administration of psilocybin completely blocked the subjective effects of alcohol. Of the remaining six participants, five reported that psilocybin had no impact on their subjective response to alcohol, while the final subject stated that psilocybin actually enhanced the subjective effects of alcohol (i.e. caused an increase in the subjective sense of intoxication). Thus, although some participants reported a tendency for psilocybin to weaken the subjective effects of alcohol, this effect was not as pronounced as that associated with LSD; while 100 per cent of LSD users reported at least

partially diminished alcohol effects, a full 40 per cent of psilocybin users failed to report any antagonizing effect. Because very little is known about the pharmacological actions of psilocybin, and the precise nature of the interaction between psilocybin and alcohol was not clearly revealed by the present data, the remainder of the discussion will focus primarily on findings concerning the concurrent administration of LSD and alcohol.

Prior to commencing a discussion of hypothesized means by which LSD could attenuate subjective alcohol effects, it would seem appropriate to address some possible limitations of the present study. The first issue concerns whether or not study participants could reliably recount their experiences with the concurrent use of alcohol and hallucinogenic drugs. There are several indications that memory is not significantly impaired by the administration of hallucinogens. For example, there are numerous reports that LSD experiences can be accurately and vividly recalled (e.g. Hofmann, 1980; Abraham *et al.*, 1996) suggesting that LSD intoxication has a negligible impact on memory storage and retrieval. The accuracy of recollections reported by subjects in the present study is further supported by the fact that most of these participants had extensive past hallucinogenic use and reported having used hallucinogens and alcohol in combination on several occasions. Thus, participants' responses to questions concerning alcohol-hallucinogen interaction effects were based on numerous experiences rather than a single event and compared to their own extensive experience with hallucinogenic drugs alone or in combination.

A second consideration concerns whether participants' recollections of alcohol effects were simply overwhelmed by the hallucinogenic properties of LSD and/or psilocybin. The observation that participants were able to reliably distinguish between LSD and psilocybin in terms of each drug's effect on the subjective alcohol experience when taken concurrently argues against this possibility. In addition, participants were highly consistent in their reporting concerning both the nature and strength of alcohol-drug interactions, despite their differential drug histories and being blind to the study's hypotheses. Finally, the amount of alcohol reportedly consumed by some participants while under the influence of LSD argues against the possibility that alcohol effects were simply overwhelmed by hallucinogen intoxication. For example, participant M described an occasion on which he had 24 drinks following the use of LSD and T recalled

drinking 18 beers, a half bottle of liqueur and 13 ounces of vodka while in a state of LSD intoxication.

While both the consistency and magnitude of reports of LSD's effect on alcohol intoxication attest to its reliability, it is important to note that in any retrospective study certain issues of potential bias may arise. Because subjects were not randomly selected it remains possible that their experiences are not characteristic of all hallucinogenic drug users. However, this possibility was minimized by the fact that participants were recruited from four different 'communities' of drug users through snow-ball sampling. It should also be noted that several additional steps were taken to avoid possible interviewer bias. First, questions used for queries about potential drug interactions were standardized and neutrally worded. Second, three different interviewers were used and all interviews were blindly rated. Finally, raters were instructed to eliminate cases where interviewer bias was present.

The present data are consistent with the notion that there is a direct, pharmacological interaction between LSD and alcohol. Although the current methodology did not directly examine either of these drugs' pharmacological effects, LSD is known to act at various serotonergic and dopaminergic receptor types (e.g. Peroutka, 1994; Watts *et al.*, 1995) implicated in ethanol reinforcement (e.g. Koob *et al.*, 1998) and discriminative stimulus (e.g. Kostowski and Bienkowski, 1999). This raises the possibility that LSD's effect on the subjective response to alcohol may involve an action at one or more of these systems.

LSD exerts effects at several 5-HT receptor subtypes, displaying affinity for 5-HT_{1A}, 5-HT_{1B}, 5-HT_{1D}, 5-HT_{1E}, 5-HT_{2A}, 5-HT_{2C}, 5-HT_{5A}, 5-HT_{5B}, 5-HT₆, and 5-HT₇ (Peroutka, 1994; Bonson *et al.*, 1996). With respect to LSD's multiple 5-HT actions, activity at 5-HT_{2C} and/or 5-HT_{1B} receptors would seem to represent the best candidates for LSD's attenuation of subjective alcohol effects, as these are the only 5-HT receptor subtypes that have been directly implicated in the formation of the ethanol cue (Maurel *et al.*, 1998; Wilson *et al.*, 1998; Kostowski and Bienkowski, 1999). Mixed 5-HT_{1B/2C} receptor agonists completely generalize to the ethanol cue in rats (Kostowski and Bienkowski, 1999; Maurel *et al.*, 1998; Szeliga and Grant, 1998), and are known to produce ethanol-like responses in humans (Kostowski and Bienkowski, 1999). Evidence also suggests that the blockade of 5-HT_{1B} and 5-HT_{2C} receptors, alone or in combination,

interferes with alcohol's discriminative stimulus properties (Maurel *et al.*, 1998). In their review article, Kostowski and Bienkowski (1999) refer to such findings, and conclude that serotonergic neurotransmission plays a key role in the formation of the ethanol cue, and that this cue is primarily mediated through 5-HT_{1B} and 5-HT_{2C} receptors.

Unfortunately, the precise nature of LSD's action at 5-HT_{1B} remains unknown (Peroutka, 1994). The present discussion will therefore focus on the actions of LSD at 5-HT_{2C} receptors. LSD displays partial agonist effects at 5-HT_{2C} receptor sites (Glennon, 1990; Fiorella *et al.*, 1995; Egan *et al.*, 1998). Thus, although LSD displays high affinity, its agonist actions at 5-HT_{2C} are of low efficacy. Because LSD is a partial agonist at 5-HT_{2C} receptors, the precise nature of its effect at these sites will depend on available 5-HT concentrations and on the presence of other 5-HT agonists and/or antagonists (Glennon, 1990; see also Bonson *et al.*, 1996). At low 5-HT concentrations and/or in the absence of another agonist, LSD behaves as a low-efficacy 5-HT_{2C} agonist, achieving approximately 30 per cent maximal excitation (Marek and Aghajanian, 1996; Egan *et al.*, 1998). In the presence of a high efficacy agonist (i.e. at higher 5-HT concentrations), LSD exerts its own maximal effect while suppressing the receptor's responsiveness to the other agonist, thereby blocking the other agonist from achieving maximal excitation. Under such conditions, LSD may thus appear to function as an antagonist. LSD's partial agonist actions are consistent with the hypothesis that LSD may prevent the level of 5-HT_{2C} excitation required for the alcohol cue to be subjectively experienced, and this mechanism may underlie the findings of the present study.

The hypothesis that LSD attenuates the ethanol cue in humans through such serotonergic mechanisms is consistent with the present observation that this effect was diminished or absent following the administration of psilocybin. Although LSD and psilocybin are considered similar in terms of subjective effects, they differ with regard to serotonergic pharmacological properties. For example, LSD is known to bind to various 5-HT sites with a greater affinity than psilocybin (Glennon *et al.*, 1985; McKenna *et al.*, 1990). Furthermore, psilocybin is currently believed to act as a full agonist at 5-HT₂ receptors (Vollenweider *et al.*, 1998), whereas LSD acts as a partial agonist at these sites. If, as hypothesized, LSD exerts an effect on the alcohol cue via suppression of the requisite 5-HT₂ response, then one might expect psilocybin to fail

to fully attenuate the subjective effects of alcohol.

Aside from its actions at 5HT receptors, LSD has also been demonstrated to exert partial agonist effects at dopamine D₁ (Watts *et al.*, 1995) and D₂ (Giacomelli *et al.*, 1998) receptors. Midbrain dopaminergic activity is thought to be critical for the reinforcement of ethanol administration (e.g. Koob *et al.*, 1998). Although DA transmission is implicated in ethanol reinforcement, there is currently only limited evidence to suggest that DA transmission substantially contributes to ethanol's subjective effects. Blocking DA neurotransmission with preferential DA receptor antagonists does not appear to affect either the discriminative stimulus effects of alcohol in rats (Kostowski and Bienkowski, 1999) or the subjective effects of alcohol in humans (Litten, 1996). Although dopamine antagonists fail to significantly alter subjective ethanol effects, there is nevertheless some indication that dopamine agonists may attenuate aspects of the ethanol cue. For example, amphetamine, a potent DA receptor agonist, has been reported to antagonize ethanol effects in rats (Schechter, 1974) and cocaine administration has been shown to attenuate the sedative effects of alcohol in humans (e.g. Foltin *et al.*, 1993). Such evidence suggests that LSD's blockade of subjective alcohol effects may relate to its agonist activity at DA receptors. Unlike LSD, psilocybin does not display an affinity for DA receptors, although there is some evidence that it may increase DA transmission indirectly through serotonergic mechanisms (e.g. Vollenweider *et al.*, 1999). Nevertheless, the differential affinity of LSD and psilocybin for DA receptors may help to explain reported differences in the effects of LSD and psilocybin on the subjective response to alcohol.

In summary, the subjective effects of alcohol are antagonized by LSD and, to a lesser extent, by psilocybin. Although it is proposed that these effects may be at least partly mediated by serotonergic and/or dopaminergic mechanisms, actions at either of these systems are likely to influence a variety of physiological and neurochemical processes and the interdependence among these should not be underestimated. It should also be emphasized that the pharmacological profiles of alcohol and hallucinogens remain incompletely understood and more research is needed in order to fully understand the precise nature of the interactions among these drugs. This point is particularly relevant in the case of psilocybin, which has not been examined as extensively as either LSD

or alcohol. Indeed, without a fuller characterization of each of these drugs' effects, interpretation of the present results must remain largely speculative.

Despite the fact that the present analysis is unable to fully elucidate the mechanisms governing alcohol–hallucinogen interactions, the present findings are highly relevant for several reasons. First, this is, to the best of the authors' knowledge, the first documentation of a recreational drug eliminating subjective alcohol effects in humans. Further exploration of the interaction between LSD and alcohol revealed by the present findings may ultimately enhance our understanding of the mechanisms that mediate the subjective effects of ethanol. Second, evidence suggests that alcohol is often co-administered with hallucinogenic drugs (Martin *et al.*, 1996). Given the amounts of alcohol reportedly consumed by some participants in the present study while under the influence of LSD and the illustrated potential for LSD to attenuate alcohol effects, this practice may pose significant health risks. Third, this analysis demonstrates that, despite the fact that similarity between LSD and psilocybin has been assumed by some researchers (e.g. Strassman, 1992), there is likely some discordance in the pharmacological effects of these two hallucinogens. Finally, this study illustrates a distinctive and effective approach for delineating drug–alcohol interactions in humans.

REFERENCES

- Abraham HD, Aldridge AM, Gogia P. 1996. The psychopharmacology of hallucinogens. *Neuropsychopharmacology* **14**: 285–298.
- Bonson KR, Buckholtz JW, Murphy DL. 1996. Chronic administration of serotonergic antidepressants attenuates the subjective effects of LSD in humans. *Neuropsychopharmacology* **14**: 425–436.
- Egan CT, Herrick-Davis CT, Miller K, Glennon RA, Teitler M. 1998. Agonist activity of LSD and lisuride at cloned 5-HT_{2A} and 5-HT_{2C} receptors. *Psychopharmacol* **136**: 409–414.
- Fiorella D, Rabin RA, Winter JC. 1995. Role of 5-HT_{2A} and 5-HT_{2C} receptors in the stimulus effects of hallucinogenic drugs II: reassessment of LSD false positives. *Psychopharmacology* **121**: 357–363.
- Foltin RW, Fichman MW, Pippen PA, Kelly TH. 1993. Behavioral effects of cocaine alone and in combination with ethanol or marijuana in humans. *Drug Alcohol Depend* **32**: 93–106.
- Giacomelli S, Palmery M, Romanelli L, Cheng CY, Silvestrini B. 1998. Lysergic acid diethylamide (LSD) is a partial agonist of D₂ dopaminergic receptors and it

- potentiates dopamine-mediated prolactin secretion in lactotrophs *in vitro*. *Life Sci* **63**: 215–222.
- Glennon RA. 1990. Do classical hallucinogens act as 5-HT₂ agonists or antagonists? *Neuropsychopharmacology* **3**: 509–517.
- Glennon RA, Titeler M, McKenney JD. 1985. Evidence for 5-HT₂ involvement in the mechanism of action of hallucinogenic agents. *Life Sci* **35**: 2505–2511.
- Hofmann A. 1980. *LSD: My Problem Child*. McGraw-Hill: New York.
- Jackson W. 1997. *Methods: Doing Social Research*. Prentice Hall: Scarborough.
- Koob GF, Roberts AJ, Schultes G, Parsons LH, Heyser CJ, Hyytia P, Merlo-Pich E, Weiss F. 1998. Neurocircuitry targets in ethanol reward and dependence. *Alcohol Clin Exp Res* **22**: 3–9.
- Kostowski W, Bienkowski P. 1999. Discriminative stimulus effects of ethanol: neuropharmacological characterization. *Alcohol* **17**: 63–80.
- Litten RZ. 1996. Pharmacotherapies for alcohol problems: a review of research with focus on development since 1991. *Alcohol Clin Exp Res* **20**: 859–876.
- Mangini M. 1998. Treatment of alcoholism using psychedelic drugs: a review of the program of research. *J Psychoactive Drugs* **30**: 381–418.
- Marek GJ, Aghajanian GK. 1996. LSD and the phenethylamine hallucinogen DOI are potent partial agonists at 5-HT_{2A} receptors on interneurons in rat piriform cortex. *J Pharmacol Exp Ther* **278**: 1373–1382.
- Martin CS, Kaczynski NA, Maaisto SA, Tarter RE. 1996. Polydrug use in adolescent drinkers with and without DSM-IV alcohol abuse and dependence. *Alcohol Clin Exp Res* **20**: 1099–1108.
- Maurel S, Schreiber R, De Vry J. 1998. Role of 5-HT_{1B}, 5-HT_{2A} and 5-HT_{2C} receptors in the generalization of 5-HT receptor agonists to the ethanol cue in the rat. *Behav Pharmacol* **9**: 337–343.
- McKenna DJ, Repke DB, Lo L, Peroutka SJ. 1990. Differential interactions of indolealkylamines with 5-hydroxytryptamine receptor subtypes. *Neuropharmacology* **29**: 193–198.
- Peroutka SJ. 1994. 5-Hydroxytryptamine receptor interactions of D-lysergic acid diethylamide. In *50 Years of LSD: Current Status and Perspectives of Hallucinogens*, Plescher A, Ladewig D (eds). Parthenon: New York; 19–29.
- Schechter MD. 1974. Effects of propranolol, d-amphetamine and caffeine on ethanol as a discriminative cue. *Eur J Pharmacol* **29**: 52–57.
- Strassman RJ. 1992. Human hallucinogen interactions with drugs affecting serotonergic neurotransmission. *Neuropsychopharmacology* **7**: 241–243.
- Szeliga KT, Grant KA. 1998. Analysis of the 5-HT₂ receptor ligands dimethoxy-4-indophenyl-2-amino-propane and ketanserin in ethanol discriminations. *Alcohol Clin Exp Res* **22**: 646–651.
- Vollenweider FX, Vollenweider-Scherpenhuyzen MFI, Babler A, Vogel H, Hell D. 1998. Psilocybin induces schizophrenia-like psychosis in humans via a serotonin-2 agonist action. *NeuroReport* **9**: 3897–3902.
- Vollenweider FX, Vontobel P, Hell D, Leenders KL. 1999. 5-HT modulation of dopamine release in basal ganglia in psilocybin-induced psychosis in man — a PET study with [¹¹C]raclopride. *Neuropsychopharmacology* **20**: 424–433.
- Watts V, Lawler CP, Fox DR, Neve KA, Nichols DE, Mailman RB. 1995. LSD and structural analogs: pharmacological evaluation at D₁ dopamine receptors. *Psychopharmacology* **118**: 401–409.
- Wilson AW, Neill JC, Costall B. 1998. An investigation in to the effects of 5-HT agonists and receptor antagonists on ethanol self-administration in the rat. *Alcohol* **16**: 249–270.

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